

Python Cheat Sheet: Lists, Tuples, Dictionaries, and Sets

1. Lists

- Ordered, mutable, allows duplicate values

Creating Lists:

```
my_list = [1, 2, 3, 4, 5]
empty_list = []
list_from_tuple = list((1, 2, 3))
mixed_list = [1, "hello", 3.14, True]
```

List Functions & Methods:

```
my_list.append(6)    # Add element at the end
my_list.insert(2, 99) # Insert at index 2
my_list.extend([7, 8]) # Extend list

my_list.remove(3)    # Remove first occurrence of 3
element = my_list.pop() # Remove last element and return it
element = my_list.pop(1) # Remove element at index 1

del my_list[2]       # Delete element at index 2
my_list.clear()      # Clear all elements

length = len(my_list) # Get length of list
index = my_list.index(2) # Get index of first occurrence of 2
count = my_list.count(3) # Count occurrences of 3

sorted_list = sorted(my_list) # Returns sorted list
my_list.sort(reverse=True)    # Sorts in place

my_list.reverse()    # Reverse the list
copied_list = my_list.copy() # Copy list
```

List Comprehension:

```
squared = [x**2 for x in range(5)]  
even_numbers = [x for x in my_list if x % 2 == 0]
```

2. Tuples

- **Ordered, immutable, allows duplicate values**

Creating Tuples:

```
my_tuple = (1, 2, 3, 4, 5)  
empty_tuple = ()  
single_element = (1,) # Single element tuple  
converted_tuple = tuple([1, 2, 3])
```

Tuple Functions & Methods:

```
length = len(my_tuple) # Get length  
index = my_tuple.index(3) # Get index of first occurrence of 3  
count = my_tuple.count(2) # Count occurrences of 2
```

Tuple Unpacking:

```
a, b, c = (1, 2, 3)
```

3. Dictionaries

- **Key-value pairs, unordered, mutable, no duplicate keys**

Creating Dictionaries:

```
my_dict = {"name": "Anish", "age": 25}  
empty_dict = {}  
dict_from_tuples = dict([("a", 1), ("b", 2)])
```

Dictionary Functions & Methods:

```
my_dict["city"] = "Delhi" # Add/Modify key-value pair  
my_dict.update({"age": 26, "country": "India"}) # Update multiple keys
```

```
value = my_dict["name"] # Get value by key
value = my_dict.get("age", "Not Found") # Get with default value
```

```
keys = my_dict.keys() # Get all keys
values = my_dict.values() # Get all values
items = my_dict.items() # Get key-value pairs
```

```
my_dict.pop("age") # Remove key and return value
del my_dict["city"] # Delete key-value pair
my_dict.clear() # Clear dictionary
```

```
copied_dict = my_dict.copy() # Copy dictionary
```

Dictionary Comprehension:

```
squares = {x: x**2 for x in range(5)}
```

4. Sets

- **Unordered, mutable, no duplicate values**

Creating Sets:

```
my_set = {1, 2, 3, 4, 5}
empty_set = set()
set_from_list = set([1, 2, 3, 3, 2])
```

Set Functions & Methods:

```
my_set.add(6) # Add element
my_set.remove(3) # Remove element (raises error if not found)
my_set.discard(3) # Remove element (does not raise error)
popped = my_set.pop() # Remove and return arbitrary element
my_set.clear() # Clear all elements

union_set = my_set.union({4, 5, 6}) # Union
intersection_set = my_set.intersection({3, 4, 5}) # Intersection
difference_set = my_set.difference({3, 4}) # Difference
symmetric_diff = my_set.symmetric_difference({3, 4}) # Symmetric difference

subset_check = my_set.issubset({1, 2, 3, 4, 5, 6}) # Check if subset
```

```
superset_check = my_set.issuperset({1, 2}) # Check if superset
```

Set Comprehension:

```
squared_set = {x**2 for x in range(5)}
```

Summary Table

Data Structure	Ordered	Mutable	Duplicate Values	Key-Value Pairs
List	Yes	Yes	Yes	No
Tuple	Yes	No	Yes	No
Dictionary	No	Yes	No (Keys Unique)	Yes
Set	No	Yes	No	No

This cheat sheet provides a quick overview of lists, tuples, dictionaries, and sets in Python, including their functions and modifications.